

0.1 自然标架的运动公式

定义 0.1

采用**张量**记号, 将正则参数曲面 S 的参数记为 u^1, u^2 , 曲面参数方程为

$$\mathbf{r} = \mathbf{r}(u^1, u^2), \quad (1)$$

记切向量

$$\mathbf{r}_\alpha = \frac{\partial \mathbf{r}(u^1, u^2)}{\partial u^\alpha}, \quad \alpha = 1, 2. \quad (2)$$

引入 **Einstein 和式约定**: 若单项式中同一指标字母出现两次, 一次为上指标、一次为下指标, 则该式表示对该指标取值 1, 2 求和. 希腊字母 $\alpha, \beta, \gamma, \delta, \xi, \eta$ 等仅取 1, 2.

曲面的单位法向量为

$$\mathbf{n} = \frac{\mathbf{r}_1 \times \mathbf{r}_2}{|\mathbf{r}_1 \times \mathbf{r}_2|}. \quad (3)$$

曲面位置向量的微分为

$$d\mathbf{r} = \mathbf{r}_\alpha du^\alpha. \quad (4)$$

定义 0.2

定义曲面的**第一类基本量**与**第二类基本量**的张量形式:

$$\begin{cases} g_{\alpha\beta} = \mathbf{r}_\alpha \cdot \mathbf{r}_\beta, \\ b_{\alpha\beta} = \mathbf{r}_{\alpha\beta} \cdot \mathbf{n} = -\mathbf{r}_\alpha \cdot \mathbf{n}_\beta = -\mathbf{r}_\beta \cdot \mathbf{n}_\alpha, \end{cases} \quad (5)$$

其中 $\mathbf{r}_{\alpha\beta} = \frac{\partial}{\partial u^\beta} \left(\frac{\partial \mathbf{r}}{\partial u^\alpha} \right)$. 曲面的第一、第二基本形式为

$$\text{I} = g_{\alpha\beta} du^\alpha du^\beta, \quad \text{II} = b_{\alpha\beta} du^\alpha du^\beta. \quad (6)$$

记度量矩阵行列式 $g = \det(g_{\alpha\beta}) = g_{11}g_{22} - (g_{12})^2$, 第二类基本量行列式 $b = \det(b_{\alpha\beta}) = b_{11}b_{22} - (b_{12})^2$.

由于 $(g_{\alpha\beta})$ 正定, 存在逆矩阵 $(g^{\alpha\beta})$, 满足

$$g^{\alpha\beta} g_{\beta\gamma} = \delta_\gamma^\alpha = \begin{cases} 1, & \alpha = \gamma, \\ 0, & \alpha \neq \gamma, \end{cases} \quad (7)$$

其中 δ_γ^α 为 Kronecker 记号, 逆矩阵显式为

$$\begin{pmatrix} g^{11} & g^{12} \\ g^{21} & g^{22} \end{pmatrix} = \frac{1}{g} \begin{pmatrix} g_{22} & -g_{12} \\ -g_{21} & g_{11} \end{pmatrix}. \quad (8)$$

表 1: Gauss 记号与张量记号对照

Gauss 记号	u	v	r_u	r_v	E	F	G	L	M	N
张量记号	u^1	u^2	r_1	r_2	g_{11}	g_{12}	g_{22}	b_{11}	b_{12}	b_{22}

定理 0.1

曲面 S 的自然标架场 $\{\mathbf{r}; \mathbf{r}_1, \mathbf{r}_2, \mathbf{n}\}$ 满足**运动公式**:

$$\begin{cases} \frac{\partial \mathbf{r}_\alpha}{\partial u^\beta} = \Gamma_{\alpha\beta}^\gamma \mathbf{r}_\gamma + b_{\alpha\beta} \mathbf{n}, \\ \frac{\partial \mathbf{n}}{\partial u^\beta} = -b_\beta^\gamma \mathbf{r}_\gamma, \end{cases} \quad (9)$$

其中 $b_\beta^\gamma = b_{\beta\xi} g^{\xi\gamma}$ 为第二类基本量的升指标形式, $\Gamma_{\alpha\beta}^\gamma$ 为 Christoffel 记号. 第一式称为 **Gauss 公式**, 第二式称为 **Weingarten 公式**, (b_β^γ) 为 Weingarten 映射在自然基 $\{\mathbf{r}_1, \mathbf{r}_2\}$ 下的矩阵.

证明 Show/Hide Proof

由于 $\mathbf{r}_1, \mathbf{r}_2, \mathbf{n}$ 线性无关, 将 $\frac{\partial \mathbf{r}_\alpha}{\partial u^\beta}, \frac{\partial \mathbf{n}}{\partial u^\beta}$ 在该标架下展开:

$$\begin{cases} \frac{\partial \mathbf{r}_\alpha}{\partial u^\beta} = \Gamma_{\alpha\beta}^\gamma \mathbf{r}_\gamma + C_{\alpha\beta} \mathbf{n}, \\ \frac{\partial \mathbf{n}}{\partial u^\beta} = D_\beta^\gamma \mathbf{r}_\gamma + D_\beta \mathbf{n}. \end{cases}$$

两边与 $\mathbf{n}$ 作内积得 $C_{\alpha\beta} = \mathbf{r}_{\alpha\beta} \cdot \mathbf{n} = b_{\alpha\beta}$; 由 $\mathbf{n}$ 为单位向量, $\mathbf{n} \cdot \frac{\partial \mathbf{n}}{\partial u^\beta} = 0$, 得 $D_\beta = 0$; 两边与 $\mathbf{r}_\xi$ 作内积得 $D_\beta^\gamma g_{\gamma\xi} = -b_{\beta\xi}$, 故 $D_\beta^\gamma = -b_{\beta\xi} g^{\xi\gamma} = -b_\beta^\gamma$.

□

定义 0.3

定义 **Christoffel 记号**:

- (1) 下指标 Christoffel 记号: $\Gamma_{\xi\alpha\beta} = \mathbf{r}_{\alpha\beta} \cdot \mathbf{r}_\xi$;
- (2) 上指标 Christoffel 记号: $\Gamma_{\alpha\beta}^\gamma = g^{\gamma\xi} \Gamma_{\xi\alpha\beta}$.

♣

命题 0.1

Christoffel 记号满足:

$$\Gamma_{\gamma\alpha\beta} = \frac{1}{2} \left(\frac{\partial g_{\alpha\gamma}}{\partial u^\beta} + \frac{\partial g_{\gamma\beta}}{\partial u^\alpha} - \frac{\partial g_{\alpha\beta}}{\partial u^\gamma} \right), \quad (10)$$

$$\Gamma_{\alpha\beta}^\gamma = \frac{1}{2} g^{\gamma\xi} \left(\frac{\partial g_{\alpha\xi}}{\partial u^\beta} + \frac{\partial g_{\xi\beta}}{\partial u^\alpha} - \frac{\partial g_{\alpha\beta}}{\partial u^\xi} \right). \quad (11)$$

Christoffel 记号由第一类基本量及其一阶偏导数唯一确定, 且关于后两个下标对称: $\Gamma_{\alpha\beta}^\gamma = \Gamma_{\beta\alpha}^\gamma$, $\Gamma_{\xi\alpha\beta} = \Gamma_{\xi\beta\alpha}$.

♠

证明 Show/Hide Proof

对 $g_{\alpha\beta} = \mathbf{r}_\alpha \cdot \mathbf{r}_\beta$ 求偏导得 $\frac{\partial g_{\alpha\beta}}{\partial u^\gamma} = \Gamma_{\beta\alpha\gamma} + \Gamma_{\alpha\beta\gamma}$, 轮换指标得到另外两式, 联立即可解出(10), 结合指标升降关系即得(11). 由 $\mathbf{r}_{\alpha\beta} = \mathbf{r}_{\beta\alpha}$, 直接可得对称性.

□

注 将张量记号还原为 Gauss 记号 ($u^1 = u, u^2 = v, g_{11} = E, g_{12} = F, g_{22} = G$), Christoffel 记号显式为

$$\begin{cases} \Gamma_{111} = \frac{1}{2} \frac{\partial E}{\partial u}, & \Gamma_{112} = \Gamma_{121} = \frac{1}{2} \frac{\partial E}{\partial v}, \\ \Gamma_{122} = \frac{\partial F}{\partial v} - \frac{1}{2} \frac{\partial G}{\partial u}, & \Gamma_{211} = \frac{\partial F}{\partial u} - \frac{1}{2} \frac{\partial E}{\partial v}, \\ \Gamma_{212} = \Gamma_{221} = \frac{1}{2} \frac{\partial G}{\partial u}, & \Gamma_{222} = \frac{1}{2} \frac{\partial G}{\partial v}, \end{cases} \quad (12)$$

$$\begin{cases} \Gamma_{11}^1 = \frac{1}{EG - F^2} \left(\frac{G}{2} \frac{\partial E}{\partial u} + \frac{F}{2} \frac{\partial E}{\partial v} - F \frac{\partial F}{\partial u} \right), \\ \Gamma_{12}^1 = \Gamma_{21}^1 = \frac{1}{EG - F^2} \left(\frac{G}{2} \frac{\partial E}{\partial v} - \frac{F}{2} \frac{\partial G}{\partial u} \right), \\ \Gamma_{22}^1 = \frac{1}{EG - F^2} \left(G \frac{\partial F}{\partial v} - \frac{G}{2} \frac{\partial G}{\partial u} - \frac{F}{2} \frac{\partial G}{\partial v} \right), \\ \Gamma_{11}^2 = \frac{1}{EG - F^2} \left(-\frac{F}{2} \frac{\partial E}{\partial u} - \frac{E}{2} \frac{\partial E}{\partial v} + E \frac{\partial F}{\partial u} \right), \\ \Gamma_{12}^2 = \Gamma_{21}^2 = \frac{1}{EG - F^2} \left(-\frac{F}{2} \frac{\partial E}{\partial v} + \frac{E}{2} \frac{\partial G}{\partial u} \right), \\ \Gamma_{22}^2 = \frac{1}{EG - F^2} \left(-F \frac{\partial F}{\partial v} + \frac{F}{2} \frac{\partial G}{\partial u} + \frac{E}{2} \frac{\partial G}{\partial v} \right). \end{cases} \quad (13)$$

若取正交参数网 $F = 0$, 公式简化为

$$\begin{cases} \Gamma_{11}^1 = \frac{1}{2} \frac{\partial \ln E}{\partial u}, & \Gamma_{12}^1 = \Gamma_{21}^1 = \frac{1}{2} \frac{\partial \ln E}{\partial v}, \\ \Gamma_{22}^1 = -\frac{1}{2E} \frac{\partial G}{\partial u}, & \Gamma_{11}^2 = -\frac{1}{2G} \frac{\partial E}{\partial v}, \\ \Gamma_{12}^2 = \Gamma_{21}^2 = \frac{1}{2} \frac{\partial \ln G}{\partial u}, & \Gamma_{22}^2 = \frac{1}{2} \frac{\partial \ln G}{\partial v}. \end{cases} \quad (14)$$

例题 0.1 求曲面 $z = f(x, y)$ 的 Christoffel 记号.

解 Show/Hide Solution

曲面参数方程为 $\mathbf{r} = (x, y, f(x, y))$, 对应 $u^1 = x, u^2 = y$, 计算切向量与二阶偏导向量:

$$\mathbf{r}_x = (1, 0, f_x), \quad \mathbf{r}_y = (0, 1, f_y),$$

$$\mathbf{r}_{xx} = (0, 0, f_{xx}), \quad \mathbf{r}_{xy} = (0, 0, f_{xy}), \quad \mathbf{r}_{yy} = (0, 0, f_{yy}).$$

第一类基本量:

$$g_{11} = 1 + f_x^2, \quad g_{12} = f_x f_y, \quad g_{22} = 1 + f_y^2, \quad g = 1 + f_x^2 + f_y^2,$$

逆度量矩阵:

$$g^{11} = \frac{1 + f_y^2}{1 + f_x^2 + f_y^2}, \quad g^{12} = -\frac{f_x f_y}{1 + f_x^2 + f_y^2}, \quad g^{22} = \frac{1 + f_x^2}{1 + f_x^2 + f_y^2}.$$

下指标 Christoffel 记号:

$$\Gamma_{111} = f_x f_{xx}, \quad \Gamma_{112} = \Gamma_{121} = f_x f_{xy}, \quad \Gamma_{122} = f_x f_{yy},$$

$$\Gamma_{211} = f_y f_{xx}, \quad \Gamma_{212} = \Gamma_{221} = f_y f_{xy}, \quad \Gamma_{222} = f_y f_{yy}.$$

上指标 Christoffel 记号:

$$\begin{aligned} \Gamma_{11}^1 &= \frac{f_x f_{xx}}{1 + f_x^2 + f_y^2}, & \Gamma_{12}^1 &= \Gamma_{21}^1 = \frac{f_x f_{xy}}{1 + f_x^2 + f_y^2}, & \Gamma_{22}^1 &= \frac{f_x f_{yy}}{1 + f_x^2 + f_y^2}, \\ \Gamma_{11}^2 &= \frac{f_y f_{xx}}{1 + f_x^2 + f_y^2}, & \Gamma_{12}^2 &= \Gamma_{21}^2 = \frac{f_y f_{xy}}{1 + f_x^2 + f_y^2}, & \Gamma_{22}^2 &= \frac{f_y f_{yy}}{1 + f_x^2 + f_y^2}. \end{aligned}$$

□

例题 0.2 设容许的参数变换是 $u^\alpha = u^\alpha(v^1, v^2)$, 命 $a_\beta^\alpha = \frac{\partial u^\alpha}{\partial v^\beta}$, 假定 $\det(a_\beta^\alpha) > 0$. 用 $g_{\alpha\beta}, b_{\alpha\beta}$ 分别表示曲面 S 在参数系 (u^1, u^2) 下的第一类基本量和第二类基本量, 用 $\tilde{g}_{\alpha\beta}, \tilde{b}_{\alpha\beta}$ 分别表示曲面在参数系 (v^1, v^2) 下的第一类基本量和第二类基本量, 则

$$\tilde{g}_{\alpha\beta} = g_{\gamma\delta} a_\alpha^\gamma a_\beta^\delta, \quad \tilde{b}_{\alpha\beta} = b_{\gamma\delta} a_\alpha^\gamma a_\beta^\delta.$$

例题 0.3 用 $(g^{\alpha\beta})$ 表示 $(g_{\alpha\beta})$ 的逆矩阵, 用 $(\tilde{g}^{\alpha\beta})$ 表示 $(\tilde{g}_{\alpha\beta})$ 的逆矩阵, 则在上题的参数变换下有

$$g^{\alpha\beta} = \tilde{g}^{\gamma\delta} a_\gamma^\alpha a_\delta^\beta.$$

例题 0.4 如果用 $\Gamma_{\alpha\beta}^\gamma$ 表示关于 $g_{\alpha\beta}$ 的 Christoffel 记号, 用 $\tilde{\Gamma}_{\alpha\beta}^\gamma$ 表示关于 $\tilde{g}_{\alpha\beta}$ 的 Christoffel 记号, 则在第 1 题的参数变换下有

$$\tilde{\Gamma}_{\alpha\beta}^\gamma = \Gamma_{\lambda\mu}^\nu a_\alpha^\lambda a_\beta^\mu A_\nu^\gamma + \frac{\partial a_\alpha^\gamma}{\partial v^\beta} A_\nu^\gamma,$$

其中 (A_ν^γ) 是 (a_α^λ) 的逆矩阵, 即 $A_\beta^\alpha = \frac{\partial v^\alpha}{\partial u^\beta}$.

例题 0.5 曲面 S 的平均曲率 H 可以表示成

$$H = \frac{1}{2} b_{\alpha\beta} g^{\alpha\beta},$$

并且 H 在第 1 题的参数变换下是不变的.

例题 0.6 下列恒等式成立:

(1)

$$g^{\alpha\delta} \Gamma_{\delta\gamma}^\beta + g^{\beta\delta} \Gamma_{\delta\gamma}^\alpha = -\frac{\partial g^{\alpha\beta}}{\partial u^\gamma};$$

(2)

$$\frac{\partial g_{\gamma\beta}}{\partial u^\alpha} - \frac{\partial g_{\alpha\gamma}}{\partial u^\beta} = g_{\beta\lambda}\Gamma_{\gamma\alpha}^\lambda - g_{\alpha\lambda}\Gamma_{\gamma\beta}^\lambda;$$

(3)

$$\Gamma_{\alpha\beta}^\beta = \frac{1}{2} \frac{\partial \ln g}{\partial u^\alpha},$$

其中 $g = g_{11}g_{22} - (g_{12})^2$.